Frame Review: Keep the Lesson

Keep the progress. Calm the blink.

A teacher has a finished recording and no editable project. During a reading pause, a caret blinks beside changing package progress. Frame Review lets an editor hold one observed caret state while the teaching content keeps its prepared timeline. The creative focus is proving what must continue, then making that proof easy to review.



Try the default real package-progress example. Inspect both observed states. Load its three editable teaching areas, check the plan, explicitly tick review and generate. Expand the live-progress proof, inspect all cuts and download the comparison. Suggestions alone never approve anything.

Vitaly Zdanevich, CC0 as listed; original seconds 62-74 prepared to 640x360/30 fps. Silent source stays silent. No visual defect added. Exact comparisons use the prepared source, not unchanged original-resolution picture.

Make the teaching timeline inspectable

OpenCV 5 temporal image differences and connected components propose stationary two-state regions; a creator determines whether a blink carries teaching information. The frozen source-matched cut policy shortens a hold rather than allowing a worse encoded return. No detector threshold, source mask, preservation tolerance or cut gate changed for this release.

The Gentoo caret [370,330,6,10] is held at inclusive frames 61-173 from observed frame 86. Selected activity falls 4 to 0 inside the hold; whole-excerpt activity falls 13 to 9 with zero new threshold event times. Entry, return and loop selected/influence RGB MAE, RMS and maximum plus the selected-mean diagnostic must be no worse than source. Existing strong source cuts can remain; this does not establish smoothness.

The progress crop [0,326,360,18] keeps 32 of 112 changing adjacent frame pairs inside the hold. Every one of its 360 frame-indexed RGB crops matches exactly. Commands keep 15 changing pairs; lower margin one. A frozen or reordered teaching sequence fails even if its total count happens to match. Counts include compression variation, not semantic progress ticks.

Gate	Fresh v7 result
Pinned Linux verification	14 stages pass; 137 backend tests; 97 studio tests, one platform skip
Actual original-source HTTP regression	All nine selected originals pass preparation, explicit plan/export, source recovery and independent native/RGB/audio/timing/cut checks
Teaching proof integrity	All actual HTTP teaching PNG hashes and decoded crops match. Mutated PNG and receipt refuse, then restore in all nine
Guided local release browser	Gentoo and Jupyter: separate load/plan/consent; ten actual artifact downloads, 24 boundary PNGs, 30 teaching PNGs, reload and actual 375/890 widths
Identity boundary	Three known prepared examples: exact source SHA plus candidate geometry required; unknown source or overlap has no guide; existing custom fields are not overwritten

Runtime manifest: dbddf025b7caf643a3b8ba95014af7d84cb4a21fbe8dd1bdf3943824fd2455ca (113 files). Strict editing, teaching, boundary and media engines are byte-identical to v6. The new guide is an editable known-example template, not arbitrary semantic teaching recognition. Initial static-image route-contract failure and the exact corrective route are retained. A damaged build cache required a clean rebuild; dependency hashes and FFmpeg pinning were not relaxed.



Nine selected sources; negatives stay visible

Real lesson	Original start	Whole source > edit	Held source > edit
jupyter	301.0 s	24 > 18	4 > 0
npm-packages	29.0 s	19 > 13	5 > 0
npm-basics	232.0 s	17 > 11	5 > 0
wikilink	60.0 s	9 > 3	4 > 0
wiki-citations	24.0 s	6 > 2	4 > 0
terminal	68.0 s	15 > 9	4 > 0
gentoo-equery	62.0 s	13 > 9	4 > 0
libreoffice-writer	150.0 s	15 > 9	6 > 0
wiki-userpage	236.0 s	17 > 11	5 > 0

Activity counts adjacent selected-box mean RGB changes of at least eight codes. All nine pass zero newly timed threshold crossings, actual encoded cut checks and complete teaching-frame checks. This release adds zero new original-source qualifications. Guided and AWS repeats are repetitions, not additional sources or users.

Earlier v6 screening retained 11 additional originals and 33 fixed windows, two additional accepted recordings, no-proposal outcomes and declined rapid-change candidates. The first seven-source/21-window screen ran on frozen v5; the second four/12 was an exploratory follow-up. Three actual command-menu/title-letter/dataset overlap plans refuse with exact source recovery and withheld edit. The first overlong Calc label request remains recorded beside its corrected harness retry. Three Toybox recordings share an author; nine cases are not nine independent creators.

The prior six-source, same-72-second comparison remains source / untuned local 5x5 sigma2 blur / older hold / matched hold = 90 / 91 / 54 / 56. Matched holds remove 34 events (37.8%) without introducing new event times; the prior longer hold introduces events in five cases. Holds total 17.6 versus 24 seconds. This does not compare against a professional editor or establish creator benefit. Full traces and all earlier failures remain versioned in v5/v6 evidence.

Exact original URLs, listed authors/licenses, original and prepared hashes and adaptations are in evidence/natural-v6/sources.json. Prepared guided sample provenance is downloadable in the app. The Writer and Portuguese Wikipedia comparisons plus credits remain in docs/natural-v6/examples; share-alike obligations apply to the Wikipedia adaptation. Imported metadata is not independent clearance or endorsement.

How this fits the OpenCV competition

Official criterion / weight	Concrete evidence and remaining gap
Technical execution / 30%	OpenCV 5 temporal/components pipeline; native/RGB/timing/audio/cut checks; actual encoded former bad Jupyter hold refused. Narrow stationary case, not a complete editor
Innovation / 20%	Creator-bounded edit plus proof that changing teaching content keeps its ordered timeline; known-source guides make that proof inspectable
Real-world impact / 20%	Finished-recording rescue task and nine selected natural cases. No measured teacher preference, time saved or learning; user study remains the largest evidence gap
User experience / 10%	Strongest real case first; one-click editable areas, separate plan and consent, mobile proofs, recoverable artifacts
Documentation / 10%	Exact source archive, original provenance, reproducible gate, selected/negative cases and source-bound report. Existing 2:46 neural-narrated video shows an earlier workflow
Cloud/reproducibility / 10%	Same ECS/Fargate service, digest-pinned release, one worker, owner-bound jobs, bounded retention. Actual cloud qualification receipt is separate from local HTTP adapter

Criterion weights were read through the official Devpost connection on 24 October 2026. This maps evidence to the rubric; it predicts no score and claims no win or optional special-award capability. The structured submission deadline is 27 October 2026, 06:45 UTC (07:45 Lagos), earlier than the overview prose, so use the earlier cutoff.

Actual AWS release: default-lesson-rescue:12. Both guided prepared-example browser flows passed, with actual artifact/PNG downloads, owner isolation, reload and downloaded 12-second playback. These repeat the local Gentoo/Jupyter cases. Playback was muted; no listening claim. Image sha256:552365d73e74e02e9fcdfc6c140e523db4165fb09abb7840dfc133cc07a526f9.

Build: docker build -t frame-review . Run fresh: docker run --name frame-review-v7-check --network none frame-review python studio/run.py --verify /tmp/qualification . Copy receipts with docker cp. Serve: docker run --rm --network host frame-review python studio/run.py --serve --port 8877 . Original-upload harness, mounting and guide checks: docs/natural-v7/REPRODUCE.md. Windows-only UI tests do not qualify media.

One 1-vCPU/2-GiB task, one worker, eight retained jobs. Supported SDR originals at most 64 MiB; selected 1-12 seconds resized to 640x360/30 fps and optional mono AAC once. Preparation can reduce legibility; exact edit checks use the prepared source. Jobs expire after 24 hours or earlier on restart. No full-lesson splice-back, moving semantic tracking or automatic publishing decision. Prior cross-host caption preparation differed and was independently checked against its actual cloud reference; no universal preparation determinism claim.

Impact hypothesis: rescue a small reading pause without re-recording the lesson. Current evidence is engineering preservation and reviewability. Next: consenting teacher/editor tasks, blinded original/edit preference and timed rescue attempts. Preference, time saved, attention, comprehension, accessibility, adoption and safety have not been measured. Real voice is deferred.